

- Q.27 Describe Bergius process for production of synthetic fuel from coal
- Q.28 Enlist important characteristics of fuel which makes them efficient.
- Q.29 Explain refining and distillation of crude oil with neat sketch.
- Q.30 Explain manufacturing of water gas and states its uses.
- Q.31 Explain working of a blast furnace in brief.
- Q.32 Describe flat plate solar collector.
- Q.33 Explain bio gas plant with neat sketch.
- Q.34 Write advantages of renewable energy.
- Q.35 Explain wind energy briefly.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the construction and working of nuclear power plant with its advantage and disadvantage.
- Q.37 Describe about origin of petroleum and explain advantage of petroleum over solid fuels.
- Q.38 Explain working principle of continuous rotary kiln used in cement industries with neat sketch.

(60)

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4th Sem / Chemical Engineering Sub : Energy Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The main component of biogas is
a) Nitrogen b) Carbon dioxide
c) Methane d) Hydrogen
- Q.2 Which fuel is commonly used in thermal power plants?
a) Coal b) LPG
c) Petrol d) Diesel
- Q.3 Calorific value of fuel means
a) Fire point
b) Ignition temperature
c) Flash point
d) Amount of heat produced on complete burning
- Q.4 Crude oil is separated by
a) Simple distillation
b) Fractional distillation
c) Crystallization
d) Filtration

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- Q.5 LPG is stored mainly in
- Solid form
 - Liquid form
 - Gaseous form
 - Semi-Solid form
- Q.6 Producer gas mainly contains
- CO and Nitrogen
 - CO₂ and Oxygen
 - Hydrogen and Oxygen
 - Nitrogen and methane
- Q.7 Solar energy reaches earth in the form of
- Infrared rays
 - UV rays
 - Electromagnetic waves
 - Sound waves
- Q.8 Which is a renewable energy source?
- Coal
 - Petroleum
 - Wind
 - Diesel
- Q.9 Coal washing mainly removes
- Moisture Content
 - Ash content
 - Carbon
 - Hydrogen
- Q.10 In hydro power plant, electricity is produced using
- Heat energy
 - Chemical energy
 - Potential and kinetic energy
 - Nuclear energy

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define renewable energy
- Q.12 Write one example of solid fuel.
- Q.13 Define calorific value.
- Q.14 Name the instrument used to measure calorific value.
- Q.15 Write any one theory of coal origin.
- Q.16 Define biomass.
- Q.17 Expand LPG.
- Q.18 Name any one petroleum product.
- Q.19 Define tidal energy.
- Q.20 Write one advantage of natural gas.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write about concept of origin of coal in brief.
- Q.22 Explain in brief proximate analysis of coal.
- Q.23 Define carbonization and Explain its types in brief.
- Q.24 Give different classifications of furnaces in brief.
- Q.25 Write importance of non-renewable source of energy for sustainable development.
- Q.26 Enlist any five petroleum products and state their uses.